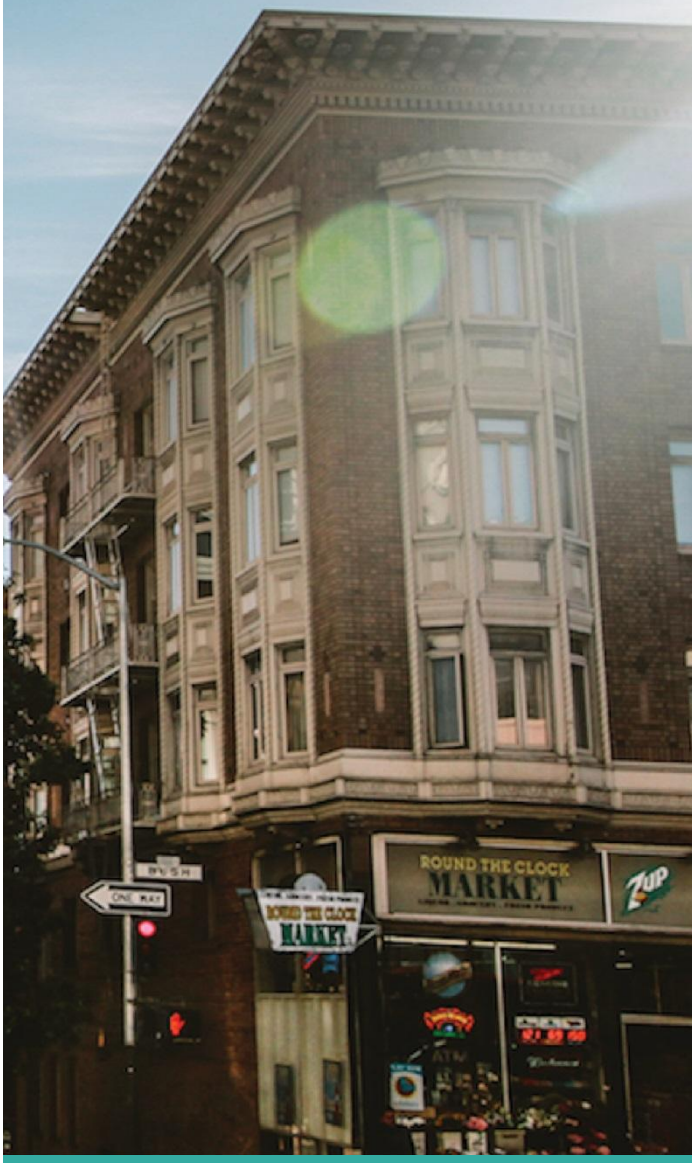
ASSIGNMENT 2
LAB 5



U26AI103

NIHIT BHAGWAT
AI BTECH DIV-B

1. A

AB

ABC

ABCD

ABCDE

looping1.c

IOP_ASSIGNMENT > iop_assignment_2 > looping1.c > main()

```c
#include<stdio.h>
int main(){
    char i;
    char j;
    for(i='A'; i<='E'; i++){
        for(j='A'; j<=i; j++){
            printf("%c ", j);
        }
        printf("\n");
    }
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2\"

A
A B
A B C
A B C D
A B C D E
PS C:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2>
```

2. 1
   12
   123
   1234
   12345

IOP_ASSIGNMENT > iop_assignment_2 > C looping2.c > main()

```c
#include <stdio.h>
int main() {
    int i, j, n = 5;

    for (i = 1; i <= n; i++) {
        for (j = n-1; j>=i; j--) {
            printf("  ");
        }
        for (j = 1; j <= i; j++) {
            printf("%d ", j);
        }
        printf("\n");
    }
    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2\" ;

        1
      1 2
    1 2 3
  1 2 3 4
1 2 3 4 5
PS C:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2>
```

3.

```
    *
   * *
  * * *
 * * * *
* * * * *
```

IOP_ASSIGNMENT > iop_assignment_2 > C 3_looping.c > main()

```c
#include<stdio.h>
int main(){
    int i, j, k, m, n=4;
    for(i=1; i<=4; i++){
        for(j=3; j>=i; j--){
            printf("  ");
        }
        for(k=1; k<=(2*i-1); k++){
            printf("* ");
        }
    printf("\n");
    }
    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\iop_assignment_
g }
    *
   * * *
  * * * * *
* * * * * * *
PS C:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2>
```

4. A

BB

CCC

DDDD

EEEEE

looping2.c ● 3_looping.c ● looping4.c ×

IOP_ASSIGNMENT > iop_assignment_2 > looping4.c > main()

```
2
3   int main() {
4       int n = 5;
5       char ch = 'A';
6
7       for (char i = 1; i <= n; i++){
8           for (int j = 1; j <= i; j++){
9               printf("%c", ch);
10              }
11          ch++;
12          printf("\n");
13          }
14      return 0;
15  }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\iop_assignme

A
BB
CCC
DDDD
EEEEE
PS C:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2>
```

5. 10000
01000
00100
00010
00001

IOP_ASSIGNMENT > iop_assignment_2 > C looping5.c > main()

```c
#include<stdio.h>
int main(){
    int i, j, n;

    for(i=1; i<=5; i++){
        for(j=1; j<=5; j++){
            if(i!=j){
                printf("0");
                }
            if(i==j){
                printf("1");
                }
            }
        printf("\n");
        }
    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\iop_assignment_

10000
01000
00100
00010
00001
PS C:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2>
```

6.
```
    1
  1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
```

```
IOP_ASSIGNMENT > iop_assignment_2 > C 6_looping.c > main()
#include<stdio.h>
int main(){
    int i, j, k, m, n=4;
    for(i=1; i<=4; i++){
        for(j=3; j>=i; j--){
            printf("  ");
        }
        for(k=1;k<=(2*i-1);k++){
            if(k>i){
                printf("%d ", (2*i-k));
            }
            else{
                printf("%d ", k);
            }
        }
        for(m=3; m>=i; m--){
            printf("  ");
        }
    printf("\n");
    }
    return 0;
}
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2\"
      1
    1 2 1
  1 2 3 2 1
1 2 3 4 3 2 1
PS C:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2>
```

7.

```
      *
    * * *
  * * * * *
* * * * * * *
  * * * * *
    * * *
      *
```

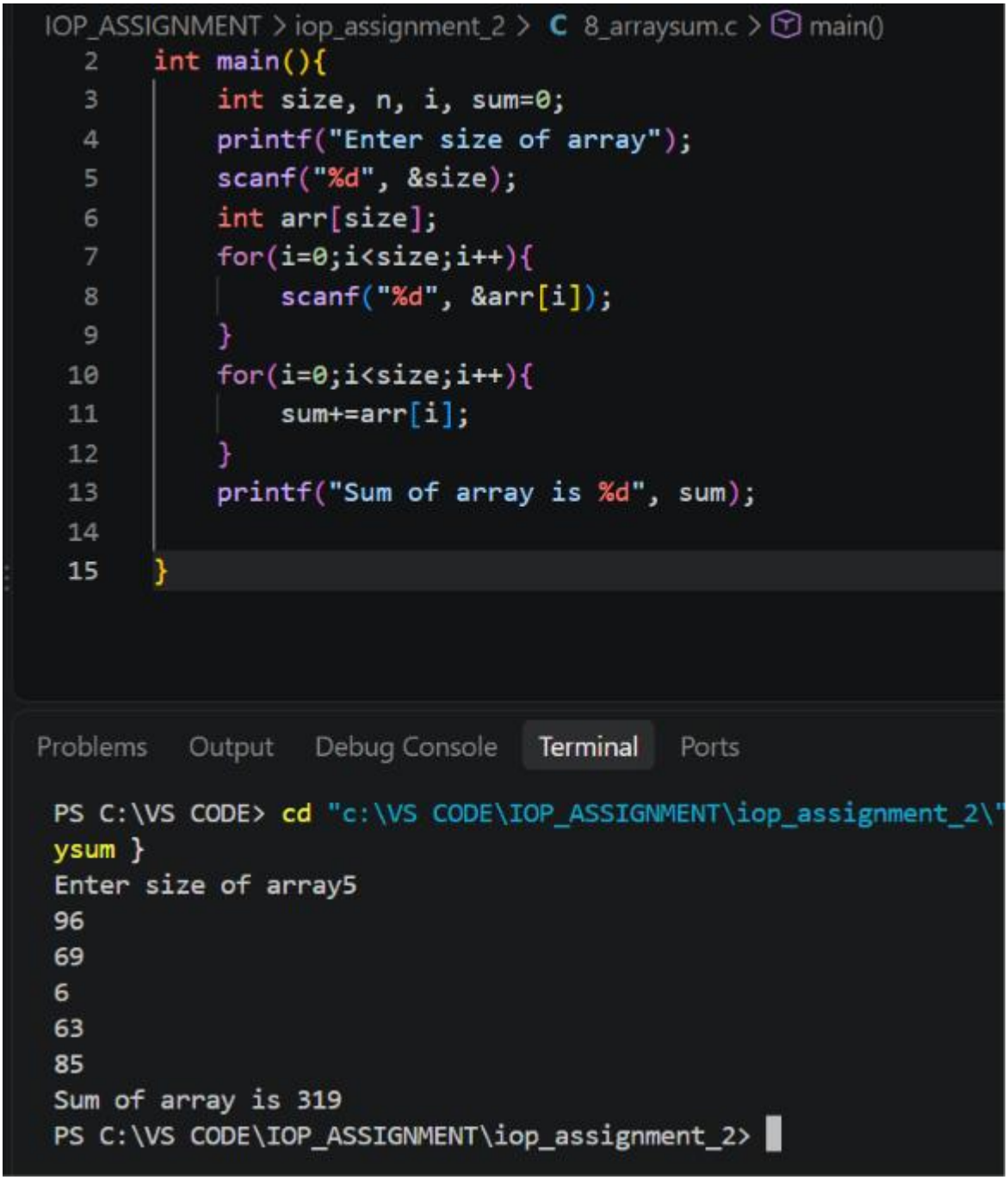
IOP_ASSIGNMENT > iop_assignment_2 > C 7_looping.c > main()

```c
#include<stdio.h>
int main(){
    int i, j ,k, n;
    for(i=1; i<=4; i++){
        for(j=3; j>=i; j--){
            printf("  ");
        }
        for(k=1; k<=(2*i-1); k++){
            printf("* ");
        }
    printf("\n");
    }
    for(i=3; i>=1; i--){
        for(j=3; j>=i; j--){
            printf("  ");
        }
        for(k=1; k<=(2*i-1); k++){
            printf("* ");
        }
    printf("\n");
    }
    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2\
      *
    * * *
  * * * * *
* * * * * * *
  * * * * *
    * * *
      *
PS C:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2>
```

8. PROGRAM TO RED AND ARRAY OF 10 INTEGER AND PRINT SUM OF NUMBERS.



9. PROGRAM TO READ AN ARRAY OF 10 INTEGER AND COUNT TOTAL NO OF ODD AND TOTAL NO. OF EVERY ELEMENT

looping2.c | 3_looping.c | 7_looping.c | 8_arraysum.c | 9_count_odd_e

IOP_ASSIGNMENT > iop_assignment_2 > 9_count_odd_even.c > main()

```
    for(i=0; i<10; i++){
        printf("%d.", i+1);
        scanf("%d", &a[i]);
    }
    for(i=0;i<10;i++){
        if(a[i]%2==0){
            printf("Number %d is EVEN\n", a[i]);
            e_count+=1;
        }
        else{
            printf("Number %d is ODD\n", a[i]);
            o_count+=1;
        }
    }
    printf("\nTotal number of EVEN no. is %d\n", e_count);
    printf("Total number of EVEN no. is %d", o_count);
}
```

Problems | Output | Debug Console | Terminal | Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2\" ; if ($?) {
Enter 10 numbers
1.9
2.6
3.3
4.8
5.5
6.2
7.7
8.4
9.1
10.89
Number 9 is ODD
Number 6 is EVEN
Number 3 is ODD
Number 8 is EVEN
Number 5 is ODD
Number 2 is EVEN
Number 7 is ODD
Number 4 is EVEN
Number 1 is ODD
Number 89 is ODD

Total number of EVEN no. is 4
Total number of EVEN no. is 6
PS C:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2>
```

10. PROGRAM TO READ AN ARRAY OF 10 INTEGER AND COUNT TOTAL NO. OF POSITIVE, NEGATIVE, AND ZERO ELEMENTS.

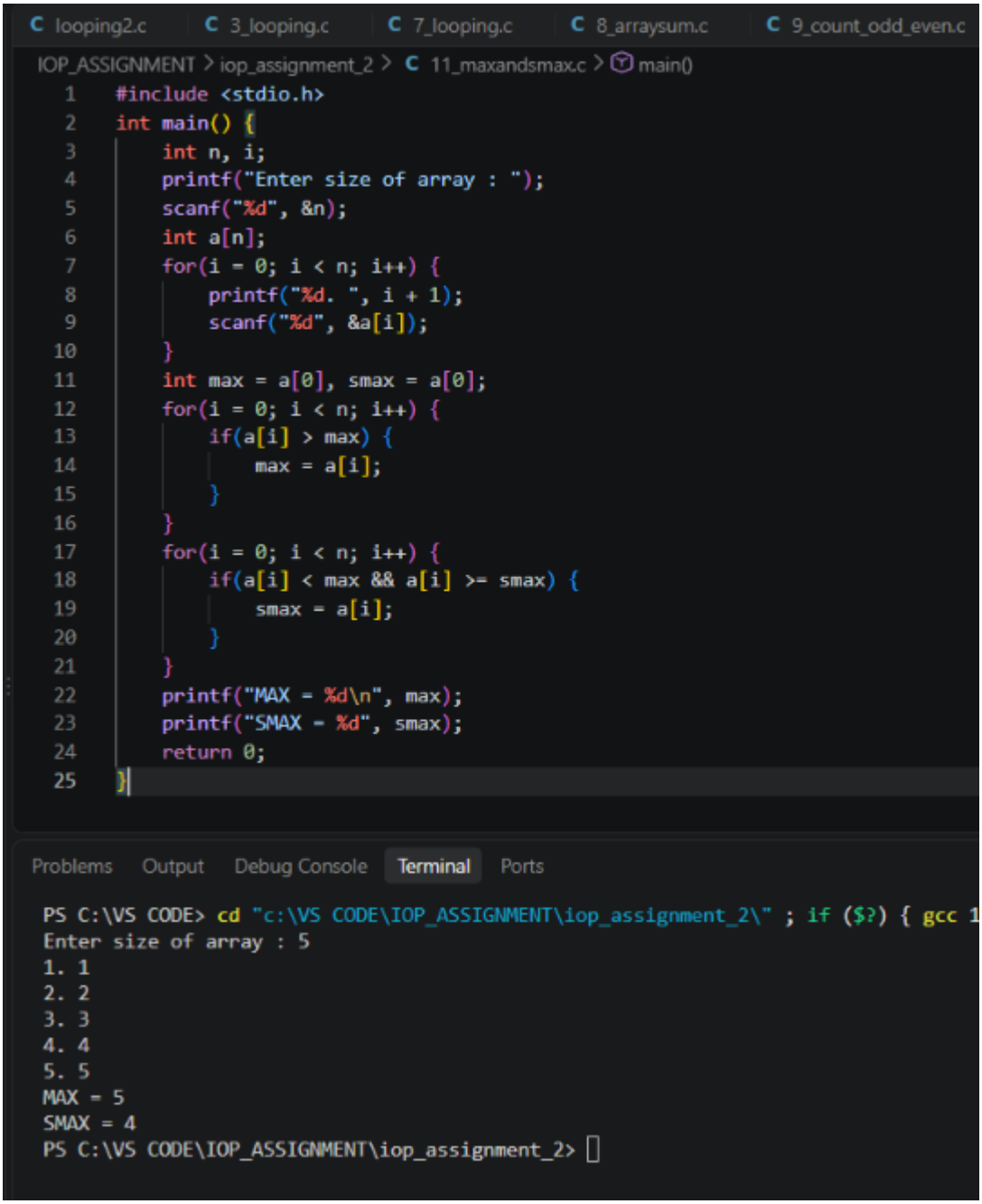
IOP_ASSIGNMENT > iop_assignment_2 > C 10_no_of+positive_zero_negative.c > main()

```c
#include<stdio.h>
int main(){
    int arr[10], i, j, pos=0, neg=0, zero=0;
    printf("Enter 10 integer\n");
    for(i=0;i<10;i++){
        printf("%d.", i);
        scanf("%d", &arr[i]);
    }
    for(i=0;i<10;i++){
        if(arr[i]>0){
            pos+=1;
        }
        if(arr[i]<0){
            neg+=1;
        }
        if(arr[i]==0){
            zero+=1;
        }
    }
    printf("The number of positive integers is %d\n", pos);
    printf("The number of negative integers is %d\n", neg);
    printf("The number of zero integers is %d\n", zero);
}
```

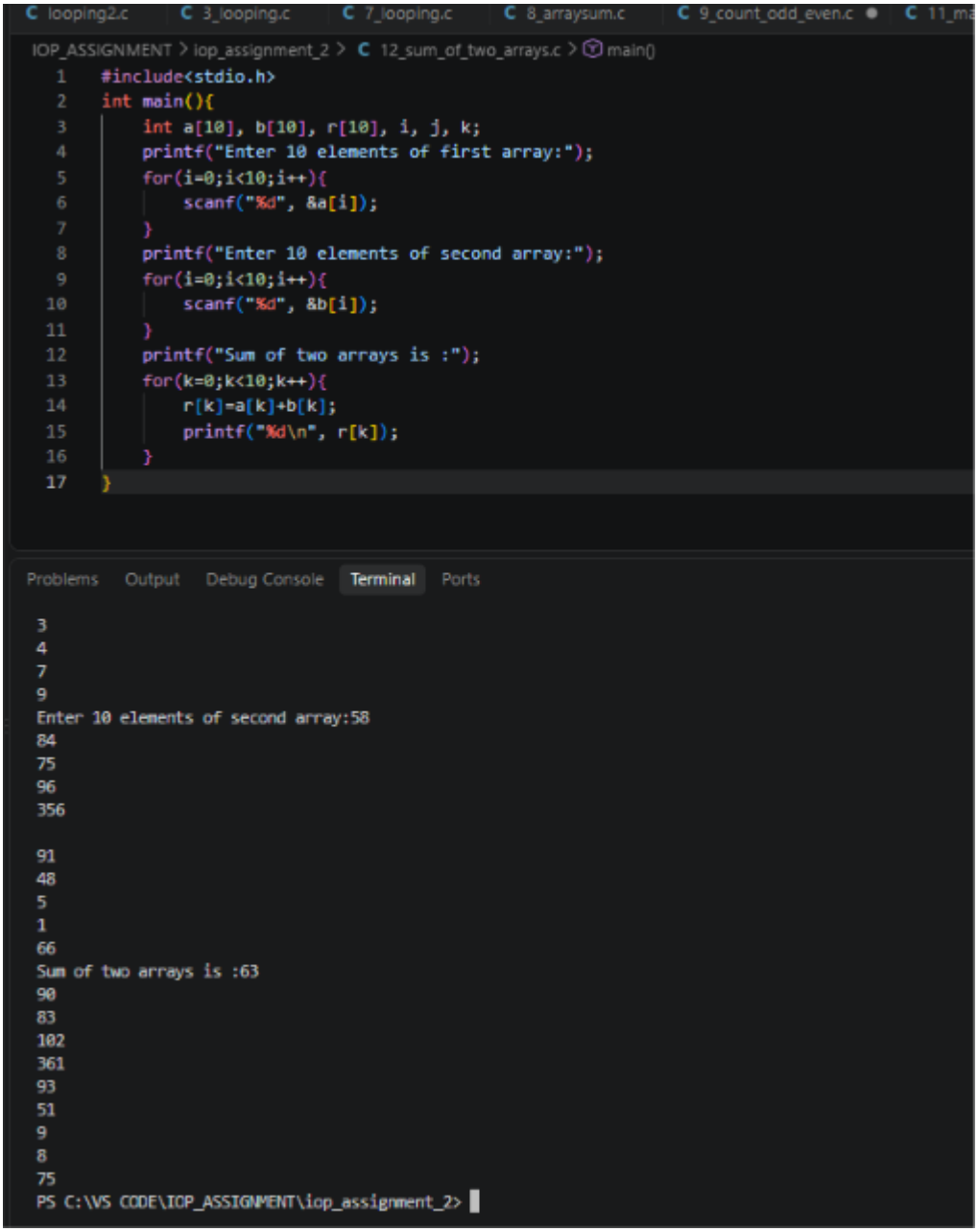
Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2\" ; if ($?) {
 }
Enter 10 integer
0.9
1.-9
2.6
3.2
4.0
5.-698
6.-123654
7.935
8.84
9.0
The number of positive integers is 5
The number of negative integers is 3
The number of zero integers is 2
PS C:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2>
```

11. PROGRAM TO READ N NOS AND FINDOUT MAX AND SECOND MAX.



12. PROGRAM TO READ TWO ARRAYS OF 10 INTEGERS AND STORE ADDATION OF THOSE ARRAYS INTO THIRD.



13. PROGRAM TO READ TWO ARRAYS OF 10 INTEGERS AND SWAP VALUES THESE ARRAYS.

looping2.c | 3_looping.c | 7_looping.c | 8_arraysum.c | 9_count_odd_even.c | 11_maxandsm

IOP_ASSIGNMENT > iop_assignment_2 > 13_array_swapping.c > main()

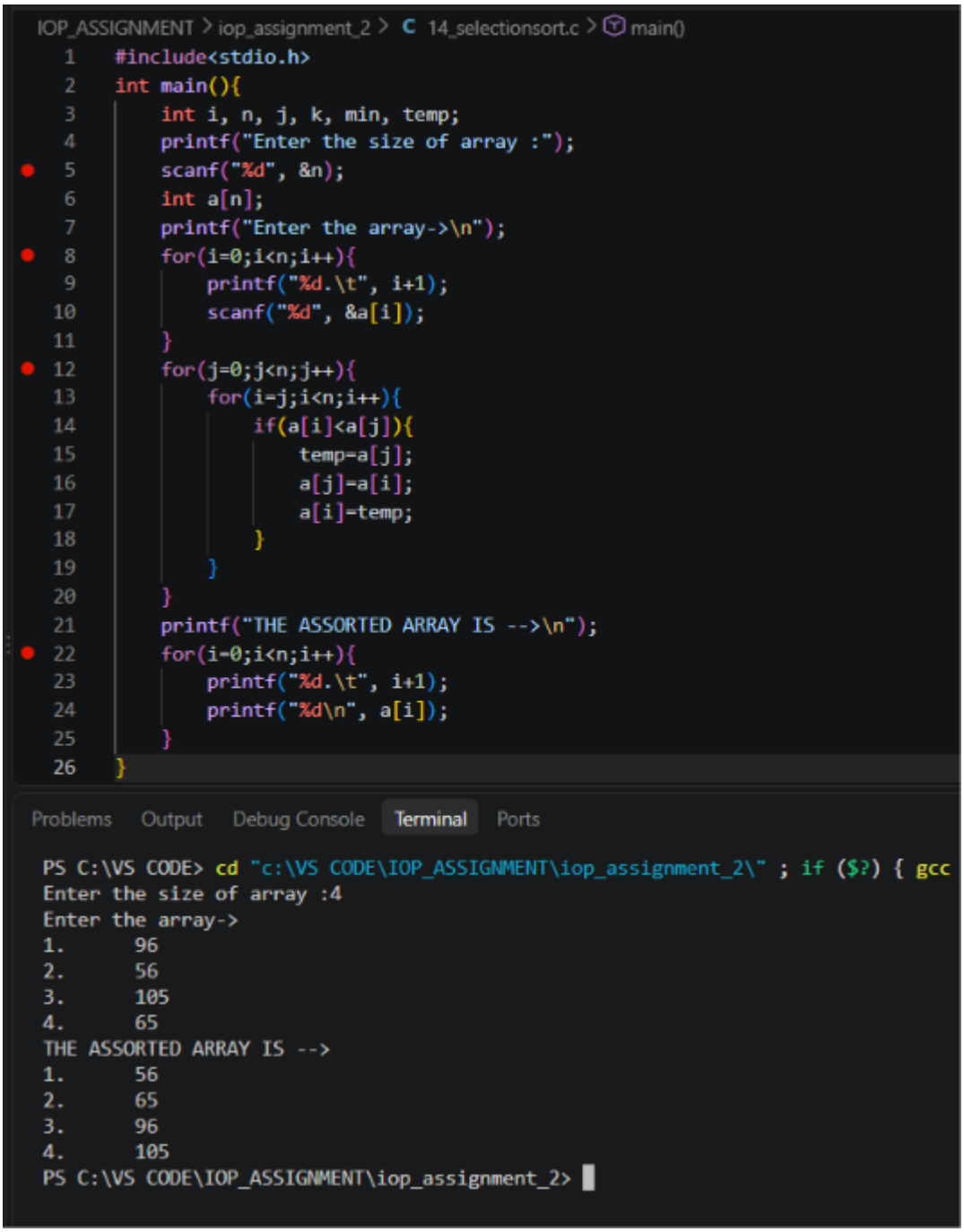
```c
#include<stdio.h>
int main(){
    int a[10], b[10], temp, i, j, k;
    printf("Enter 10 elements of first array:");
    for(i=0;i<10;i++){
        scanf("%d", &a[i]);
    }
    printf("Enter 10 elements of second array:");
    for(i=0;i<10;i++){
        scanf("%d", &b[i]);
    }
    printf("After swapping\n");
    printf("A\tB\n");
    for(k=0;k<10;k++){
        temp=a[k];
        a[k]=b[k];
        b[k]=temp;
        printf("%d\t%d\n", a[k], b[k]);
    }
}
```

Problems | Output | Debug Console | Terminal | Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2\" ; if ($?) { gcc 13_array_swapping.c -o 13_ar
Enter 10 elements of first array:9
8
7

5
4
3
2
1
0
3
Enter 10 elements of second array:0
1
2
3
4
5
6
7
8
9
After swapping
A       B
0       9
1       8
2       7
3       5
4       4
5       3
6       2
7       1
8       0
9       3
```

14. PROGRAM TO SORT (SELECTION SORT) AN ARRAY.



```
IOP_ASSIGNMENT > iop_assignment_2 > C 14_selectionsort.c > main()
```

```c
#include<stdio.h>
int main(){
    int i, n, j, k, min, temp;
    printf("Enter the size of array :");
    scanf("%d", &n);
    int a[n];
    printf("Enter the array->\n");
    for(i=0;i<n;i++){
        printf("%d.\t", i+1);
        scanf("%d", &a[i]);
    }
    for(j=0;j<n;j++){
        for(i=j;i<n;i++){
            if(a[i]<a[j]){
                temp=a[j];
                a[j]=a[i];
                a[i]=temp;
            }
        }
    }
    printf("THE ASSORTED ARRAY IS -->\n");
    for(i=0;i<n;i++){
        printf("%d.\t", i+1);
        printf("%d\n", a[i]);
    }
}
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2\" ; if ($?) { gcc
Enter the size of array :4
Enter the array->
1.      96
2.      56
3.      105
4.      65
THE ASSORTED ARRAY IS -->
1.      56
2.      65
3.      96
4.      105
PS C:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2>
```

15. PROGRAM TO SORT (BUBBLE SORT) AN ARRAY.

looping2.c | 3_looping.c | 7_looping.c | 8_arraysum.c | 11_maxandsmax.c

IOP_ASSIGNMENT > iop_assignment_2 > 15_bubblesort.c > main()

```c
    int a[n];
    printf("Enter the array->\n");
    for(i=0;i<n;i++){
        printf("%d.\t", i+1);
        scanf("%d", &a[i]);
    }
    for(j=0;j<n;j++){
        for(i=0;i<n-1;i++){
            if(a[i]>a[i+1]){
                temp=a[i];
                a[i]=a[i+1];
                a[i+1]=temp;
            }
        }
    }
    printf("sorted array is -->\n");
    for(i=0;i<n;i++){
        printf("%d.\t", i+1);
        printf("%d\n", a[i]);
    }
    return 0;
}
```

Problems | Output | Debug Console | Terminal | Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2\" ; if ($?) { gcc 15
Enter the size of array :5
Enter the array->
1.      99
2.      105
3.      96
4.      39
5.      86
sorted array is -->
1.      39
2.      86
3.      96
4.      99
5.      105
PS C:\VS CODE\IOP_ASSIGNMENT\iop_assignment_2>
```